

Hybrid Optimization Algorithm Using N Gram Based Edit Distance

M.Priya, Dr.R.Kalpana and T.Srisupriya

Abstract—Spell-checking method is used to detect and correct the incorrect words which spelled in a document. To afford the best probable solution for corrected words first we should detect the incorrect words. The several spell correcting techniques are used after detecting the wrong words or misspelled words, to produces the best accurate correct or alternate words. Spell checking is used in word processing programs, cell phones, email programs and many other applications such as blogs and forums etc. Classification of Error correction and detection provide a better understanding of different types of techniques. There are two basic approaches to correct the spelling called edit distance approach and N-gram approach. This paper deals with a new technique called N-Gram based Edit distance (NGE) for transformation of strings. For rule generation to estimate the probabilities using Log model. The proposed method is compared with the other two approaches: Edit distance and N-Gram. The proposed method yields better result than other two approaches and the hybrid algorithm outperforms both edit distance and n gram algorithm in terms of efficiency and accuracy.

Index Terms—Edit Distance, N-Gram, and N-Gram based edit distance.

I. INTRODUCTION

Spell Checking is the function of finding and flagging erroneously spelled words in a document written in a natural language. Spell Correcting is the method of interchanging the misspelled words with the most likely intentional ones. They are lots of applications has been used such as Word processing, character recognition, optical character recognition (OCR), etc. Spelling errors can be partition into two types Real-word errors and Non-word errors. Real-word errors are those error words that are suitable words in the dictionary. Non-word errors are those error words that cannot be found in the dictionary. The three types of non-word error are typographic errors, cognitive errors, and phonetic errors. Typographic errors, it frequently occur during keyboard mis-punches (e.g. "oan" - "on"), cognitive errors, it affected by the writer's misunderstandings (e.g. "existence"-"existence"), phonetic errors, it replacing a phonetically corresponding sequence of letters (e.g. "privelege" - "privilege").

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Text similarity measures show an essential role in text related research and applications such as information retrieval, text classification, document clustering, etc. Finding the similarity in between the word is the important part of text similarity, when it is used as sentence, paragraph and document similarities for approximate string matching.

In Section II discusses literature survey. The section III and IV explain classification of error correction and error detection mechanism and its preliminaries details. The section V and VI examine a new algorithm for error correction and error detection and its experimental details. The section VII concludes.

II. LITERATURE SURVEY

Spell checking defines the several methods for error correction and spell checking. It also delivers the information about many existing spell checking systems which developed for several Indian languages. They described two techniques for spell checking is (1) Statistical technique used for N Gram Analysis (2) Dictionary lookups.

Baljeet [1] the part of error detection and spell correction techniques has been surveyed. With the support of spell checkers the Existed task associated in Punjabi and the Punjabi language is also considered. With more accuracy, the Punjabi spellchecker will be implemented by using edit-distance and dictionary lookup based technique. The Error Correction three techniques are (1) N Gram Analysis (2) Edit Distance and (3) Rule Based Approach.

Similarity between two words can be measured with edit distance function. The Levenshtein algorithm [2] uses a weighting approach it is basically used for allocate a cost of 1 to each and every single edit operation. Damerau-Levenshtein describes distance between two strings by calculating the minimum number of operations required to transform one string into the other, where an operation are insertion, deletion, or substitution of a single character, or a transposition of two adjacent characters [3].

The Hamming algorithm [4] is used to calculate the distance between two strings of same length. It is considered by finding the minimum number of substitutions required to transform string x into string y. This techniques are used only when the string of the equal length. The Longest Common Subsequence [5] is used to find out the longest common subsequence of two strings. It consists of sequence of characters, it is not sequential, that perform from left to right in a string.

Shannon [6] discussed N-Gram techniques in language processing. After that it has been used in many problems such as word prediction, spelling correction, etc. The major advantage of using n-gram method is that, it is language independent. In a spelling correction function, an n-gram is a series of n letters in a word or a string. It can be used to calculate the similarity between two strings.

N-Gram, retrieval depends on character N-grams. N-gram can be used as a representation method and then the outcome are compared using vector space model with three similarity measures. Trigrams are used to Index the Arabic documents. It is extensively used in text retrieval. Suleiman [7] learning a unique approach by matching the work of two N-gram Methods such that the contiguous N-gram and the hybrid N-grams merging contiguous and non-contiguous characters. It can be used in numerous search engines that permit the search for all the arrived words inside the same sentence without coarse their order.

Probabilistic technique [8] conferred the transformations directly by trying all the prospects such as insertion, deletion, substitution. This technique is based on dictionary. To reduces the dictionary access, the system uses pre-computed table which includes missing a letter to corrections and also it is used for checking insertion, deletion and substitution

Neural net [9] uses a hybrid approach to overcome phonetic spelling errors and the four main methods of typing errors (insertion, deletion, substitution, and transposition (double substitution)). This technique benefit a Soundex-type coding approach together with transformation rules to overcome phonetic spelling errors. Phonetic spelling errors are hard to detect and correct. And it uses an n-gram approach to overcome insertion, deletion and incorporate a Hamming Distance approach to overcome substitution and transposition errors. Rule based techniques [9] describe a spelling error correction system which, to several point, follows the reaction of the human brain. It generally comprises information of probable pronunciations of the word and also spelling without the knowledge of its context. By a number of considerations the system identifies which portion of the dictionary possibly covers the word anticipated and its efficiency and accuracy is improved. Similarity keys technique allocate a key to every dictionary word and then compare using the key figured for the non-word. Then the, result is to select the most similar key as suggestion. The method used Soundex Phonetic algorithm.

III. CLASSIFICATION OF ERROR CORRECTION AND ERROR DETECTION

In Fig.1 Spell checking process includes both error correction and error detection process. The error detection is a process it generally involves of testing to watch if an input string is a dictionary word. The error correction process is used to select the appropriate word for the given error word.

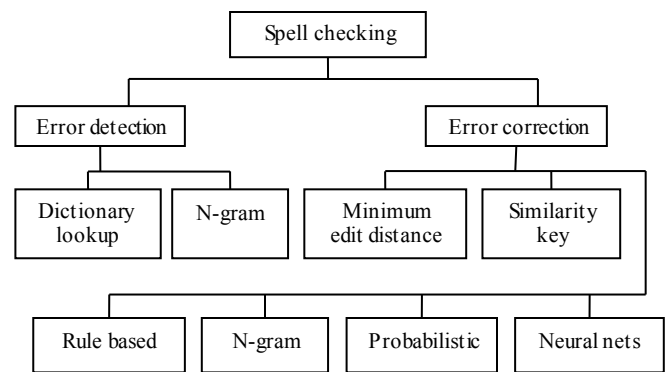


Fig. 1 classification of error correction and error detection

1) Error Detection Techniques:

a) Dictionary lookup technique

In Dictionary lookup technique, it checks each and every word of input word whether it is presence in the dictionary or not. In order that word is present in the dictionary it is right word else it is error word. Hash table is used for quick access to a dictionary. First the work is to view an input string, calculates the hash addresses and then fetch the word that are gathered on that address in the pre-designed hash table. If hash address is distinct from the input string, a wrong spelling is determined.

b) N-Gram analysis

In this, N-gram Analysis Method is used. N-grams are n-character sub arrangement of words or strings where n is set to one, two or three. One character n-grams are termed as unigrams, two character n-grams are termed as bi-grams, and three character n-grams are termed as trigrams. In n-gram detection technique by considering every one n-gram is an input string and observing in a precompiled table of n-gram statistics to its occurrence of words.

2) Error Correction Techniques

a) Minimum Edit Distance

In minimum edit distance is the use of the minimum number of operations (insertions, deletions and substitutions) recommended to transforming one text string into another. In its initial structure, minimum edit distance algorithms are used it depend upon the m similarity between incorrectly spelled string and the dictionary of m words.

There are three algorithms for computing Minimum edit distance they are Levenshtein algorithm, Hamming, Longest Common Subsequence. In the Levenshtein algorithm, the cost is set to 1 for each and every edit operation. The Hamming algorithm is to calculate the distance between two strings of equal length. In the Longest Common Subsequence algorithm is used to catch the Difference between two words.

b) Similarity key

In Similarity key techniques the task is to plan every single string into a key, there are spelled string and misspelled string, Such that spelled string gains similar keys.

Therefore when key is calculated for wrongly spelled strings then it deliver an indicator to all equivalently spelled words in the dictionary. The two most well-known methods are Soundex algorithm and the SPEEDCOP System .Soundex Algorithm is used for indexing words placed on their phonetic sound. The SPEEDCOP System are used when it automatically correcting spelling errors essentially typing errors in a very large database.

c) Rule Based Technique

In Rule Based Technique, transforming misspellings into valid words are based on rules that work to signify understanding of frequent spelling errors design in the method of rules. There are two simple methods they are transition probabilities and confusion probabilities. The methods are related to n-grams. It provides the probability of a letter followed by another letter. The probability can be assessed through computing n gram frequency for the words from a corpus. In Confusion Probabilities, they are used to measure of how frequently a known letter is incorrect for another given letter.

d) Neural networks

It is similar to before used methods, but it has to advance a little more. The present methods are built on back-propagation networks, they work by means of output node and input node, one output node on behalf of every term in the dictionary and an input node for every probable n-gram in every one position of the term, where n generally is 1 or 2. It uses Back propagation algorithm

IV. PRELIMINARIES

A .Edit Distance:

Given two strings return the “distance” between the two strings as calculated by Edit distance. The minimum number of “character edits operations” (insertion, deletion, substitution, Transposition) necessary to transforming one order into the other. This distance is a key importance in Spell checking, Speech recognition, etc. In spell checking, given a string $S_1 = [a_1 a_2 a_3 \dots a_n]$ and $S_2 = [b_1 b_2 b_3 \dots b_m]$ as the minimum cost of transforming S_1 into S_2 by means of three operations such as insert, delete, and substitution, where simply unit cost operations are considered in edit distance. If s is "football" and t is "Football" and then distance is 0, because no transformations are desired, already strings are equal. If s is "Football" and t is "Foetball", then ED is 1, because one substitution is necessary to transforms into t.

The stages to implement this technique are as follows, given the string pairs(s, t), assign n to be the length of s and assign m to be the length of t. For each character of s and t and then cost is determined by checking if $s[i]$ equals to $t[j]$ means it returns string else it is different character means it calculates the cost for each edit operation. And finally they return the string with smallest minimum edit distance. Check they are equal or not. If equal then return 0 otherwise calculate the cost of the operation. For the strings of different length, sequence or latent alignment technique is applied and then the above procedure is repeated.

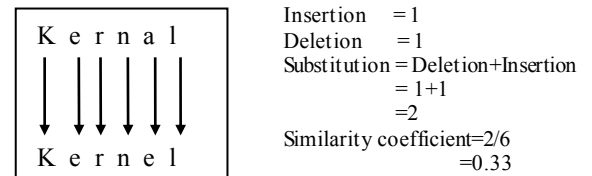


Fig.2 Edit Distance

B.N-Gram

N-gram models are extensively used in statistical natural language processing. In speech recognition, phonemes and sequences of phonemes are displayed using a n-gram .N-gram is defined as a technique to discover misspelled words in a corpus of text. A check is through an n-dimensional matrix wherever n-gram occurrences are kept. If uncommon n-gram is found the word is labelled as a misspelling, else not. An n-gram is a set of sequential characters occupied from a string with a size of whatever the 'n'. If n is 1 is a unigram, if n is 2 is a Bigram, if n is 3 is a trigram. For example, the following is a list of all distinctive character-level order-2 n-grams in the word.

From the Fig. 3 N-Gram approach divides the word into 2 grams, generally correct “kernal” made by the “kernel” using substitution operation substitute each character for creating up the word. If the “kernel” meant to e.g. press the character “e” a common mistyping is to instead press one of the characters actually close to ”r” on the keyboard, e.g. by pressing “w”, ”r”, ”d”, ”s” and so on. Afterwards the similarity coefficient are calculated by n-gram method, they retrieve both the number of identical n-grams and number of unique grams present in that two strings i.e “kernal” and “kernel”. Table displays an example for the computation of the similarity coefficient for the misspelled word “kernal” and the correct word “kernel” by means of an n-gram with n=2 (bigram).

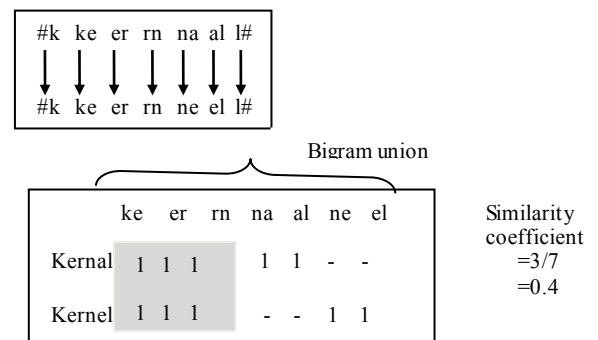


Fig.3 N-Gram

V. HYBRID ALGORITHM USING N-GRAM BASED EDIT DISTANCE

In the method, input string and output string pairs are specified as training data. From the training string pairs rules are extracted. Along with that rules weight are calculated by bayes rule. Then the rule with weight are denoted as $u \rightarrow v, w$. Here number of transformation rules are generated for the string pairs, so that it uses binary classifier to specify whether the rules are applied or not i.e the classifier '0' specifies it is not applied, 1 indicates the rule is applied. The step by step

procedures for Hybrid Algorithm using N-gram based Edit Distance are as follows.

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Get the string pairs (si, ti)
Apply Bigram for the pairs (si, ti)
Then common bigrams are predict from (si, ti)
And find the union of bigram from (si, ti)
Apply n-gram based edit distance and create
transformation rule
End

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Fig.4 shows the proposed method uses both n-gram features as well as edit distance features, from the figure it shows, first it divides the misspelled word and correct word into bigrams (as shown in the figure). From the two set, union of bigrams are predicted, so it generates many bigrams such as (ee, eg, ed, dg). Due to this reason, again to reunion the bigrams from union of bigrams, (ee → ed, eg → dg) hence the bigrams are minimized (e → d) for each word there gives rules, (p(k|c) = 0.03...etc) based on that it calculates the probabilities along with that weight are evaluated by bayes rules. Then the parameters in N-gram model can be estimated with Maximum Likelihood Estimation technique. Due to some instance zero probability occurs, to overcome this problem it uses smoothing technique, it increases the probability upward and to improve estimation and the accuracy of the model.

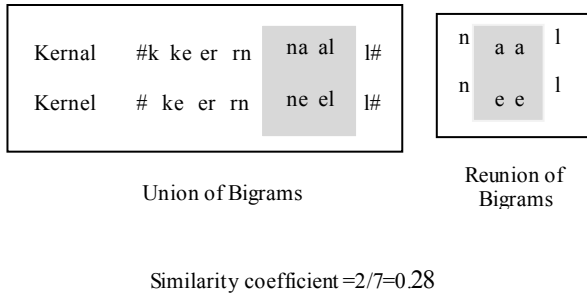


Fig.4 Hybrid Algorithm Using N-Gram Based Edit Distance

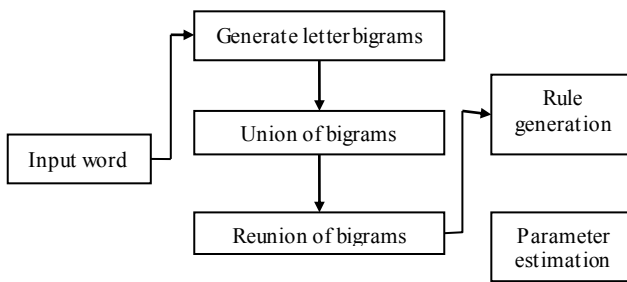


Fig.5 Workflow of Hybrid Algorithm Using N-Gram Based Edit Distance

A) MLE (Maximum likelihood Estimation)

From the training data, $P(s_i, do)$ where s_i is an input string and do is an output string ω . During transformation many number of rules are generated. We are using maximum likelihood estimation for parameter estimation.

The conditional likelihood function is defined as follows

$$L_N(\omega; d|s) = \prod_{i=1}^N f_{D_i} \quad (1)$$

Where denotes the conditional probability of given Then the log likelihood function becomes

$$L_N(\omega; d|s) = \sum_{i=1}^N f_{D_i} \quad (2)$$

VI. EXPERIMENTAL RESULTS

In the following, we display the outcomes of some experiments done on the Dict dataset. Afterwards, we compared the words with the three approaches. Then it is displaying that the proposed approach produced the best results. For the valuation, we used the Dict dataset. Initially, those dataset do not encompass any misspellings; however, for valuation purposes numerous words were changed, resulting in non-word and real-word errors in the text. Spell-checking the test data by the proposed algorithm corrected 94% of total errors were corrected successfully. Edit distance correct misspelled words of (87%), while N-gram approach can correct (90%) words.

The experimental results are shown in the Fig 6. With increasing the maximum number of rules and word length, the running time increases in edit distance approach and N-Gram approach.

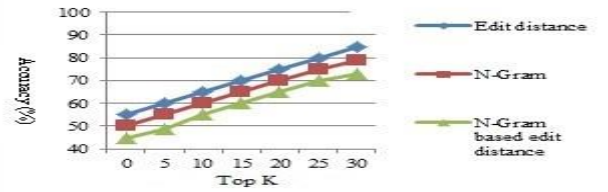


Fig.6. Accuracy Comparison between other method and Hybrid algorithm using N-Gram based Edit distance

In Fig.7 shows that running time of Hybrid algorithm using N-Gram based edit distance method is less than that of Edit Distance and N-Gram especially when the word length is short. To conclude that the N-Gram based Edit Distance is more efficient than other two methods. In Fig.8 shows that running time of Hybrid algorithm using N-Gram based edit distance method is less when compared with other two methods especially when the rule set size is small.

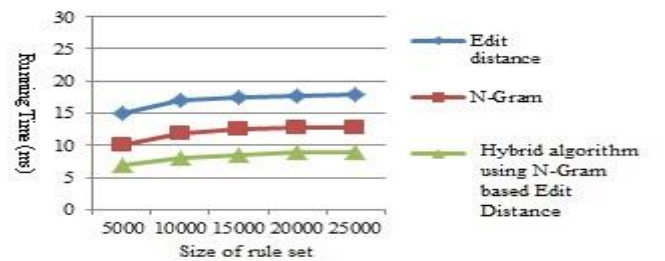


Fig.8. Efficiency Evaluation with different size of rule set

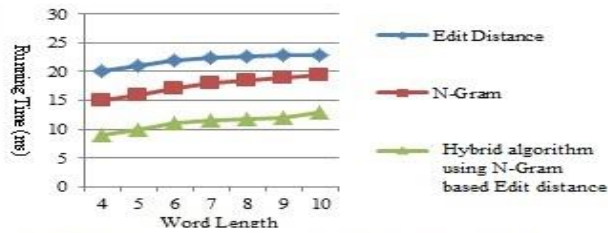


Fig. 7 Efficiency comparison between other two method and Hybrid algorithm using N-Gram based edit distance

The proposed method gives efficient result especially when length of the query is long or short. From the comparison of three approaches, our proposed method yield better result than other two approaches and normally, many rules are generated during transformation. By using a new approach Hybrid algorithm using N-Gram based Edit Distance rules are minimized. From the comparison of three approaches, our proposed method minimizes the rule when compared with other two approaches. To conclude that the method N-Gram based edit Distance is more efficient than edit distance and N-Gram.

VII. CONCLUSION

In this survey three approaches were discussed, Edit distance N-Gram and N-Gram based Edit distance. This paper illustrates hybrid algorithm using N-Gram based Edit distance method which combines edit distance approach and N-Gram approach. The existing method cannot handle rule generation and the proposed method provides the optimized rule. Furthermore, we presented evaluations on Dict dataset. The gained results outperformed other state-of-the art methods. Hybrid algorithm using N-Gram based Edit distance method is efficient and it gives good performance when compared with the other two methods.

REFERENCES

- [1] Baljeet Kaur, Review On Error Detection and Error Correction Techniques in NLP: Volume 4, Issue 6, June 2014 ISSN: 2277 128 X, International Journal of Advanced Research in Computer Science and Software Engineering
- [2] Levenshtein, V.I., "Binary codes capable of correcting deletions, insertions, and reversals", *Cybernetics and Control Theory*, 10(8), 707-710, 1966
- [3] Hall, P. A. V. & Dowling, G. R. (1980) Approximate string matching. *Comput. Surveys*, 12:381-402.
- [4] Hamming, Richard W., "Error detecting and error correcting codes", *Bell System Technical Journal* 29 (2): 147-160, 1950.
- [5] L. Allison and T.I. Dix., "A bit-string longest common-subsequence algorithm", *Information Processing Letters*, 23:305-310, 1986
- [6] C. E. Shannon, "Prediction and entropy of printed English," *Bell Sys.Tec. J.* (30):50-64, 1951
- [7] Suleiman H. Mustafa and Qasem A. Al- Radaideh, "Using N-Grams for Arabic Text Searching", *Journal of the American Society for information science and technology*, 55(11):1002-1007, 2004.
- [8] K. Church and W. A. Gale, "Probability scoring for spelling correction," *Statistics and Computing*, Vol. 1, No. 1, pp. 93-103, 1991.
- [9] V. J. Hodge and J. Austin, "A comparison of standard spell checking algorithms and novel binary neural approach," *IEEE Trans. Know. Dat. Eng.*, Vol. 15:5, pp. 1073-1081, 2003
- [10] E. J. Yannakoudakis and D. Fawthrop, "An intelligent spelling error corrector," *Information Processing and Management*, 19:1, 101-108, 1983.
- [11] Wikipedia, list of Common Misspelling Word List, http://en.wikipedia.org/wiki/Wikipedia:List_of_common_misspellings, 05.10.2006.
- [12] Neha Gupta, Pratistha Mathur, Spell Checking Techniques in NLP: A Survey, Volume 2, Issue 12, December 2012, ISSN: 2277 128X, International Journal of Advanced Research in Computer Science and Software Engineering
- [13] Holmes. D and McCabe. Improving precision and recall for soundex retrieval. In *Information Technology: Coding and Computing, 2002 Proceedings*. International Conference on, pages 22-26.
- [14] A. Arasu, S. Chaudhuri, and R. Kaushik, "Learning string transformations from examples," *Proc. VLDB Endow.*, vol. 2, pp. 514-525, August 2009
- [15] S. Tejada, C. A. Knoblock, and S. Minton, "Learning domain independent string transformation weights for high accuracy object identification," in *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, ser. KDD '02*. New York, NY, USA: ACM, 2002, pp. 350-359.
- [16] M. Hadjieleftheriou and C. Li, "Efficient approximate search on string collections," *Proc. VLDB Endow.*, vol. 2, pp. 1660-1661, August 2009.
- [17] C. Li, B. Wang, and X. Yang, "Vgram: improving performance of approximate queries on string collections using variable-length grams," in *Proceedings of the 33rd international conference on Very large data bases, ser. VLDB '07*. VLDB Endowment, 2007, pp. 303-314
- [18] X. Yang, B. Wang, and C. Li, "Cost-based variable-length-gram selection for string collections to support approximate queries efficiently," in *Proceedings of the 2008 ACM SIGMOD international conference on Management of data, ser. SIGMOD '08*. New York, NY, USA: ACM, 2008, pp. 353-364.
- [19] M. Li, Y. Zhang, M. Zhu, and M. Zhou, "Exploring distributional similarity based models for query spelling correction," in *Proceedings of the 21st International Conference on Computational Linguistics and the 44th annual meeting of the Association for Computational Linguistics, ser. ACL '06*. Morristown, NJ, USA: Association for Computational Linguistics, 2006, pp. 1025-1032
- [20] A. R. Golding and D. Roth, "A winnow-based approach to context-sensitive spelling correction," *Mach. Learn.*, vol. 34, pp. 107-130, February 1999.

- [21] J. Guo, G. Xu, H. Li, and X. Cheng, “A unified and discriminative model for query refinement,” in Proceedings of the 31st annual international *ACM SIGIR* conference on Research and development in information retrieval, ser. SIGIR '08. New York, NY, USA: ACM, 2008, pp. 379–386.
- [22] A. Behm, S. Ji, C. Li, and J. Lu, “Space-constrained gram-based indexing for efficient approximate string search,” in Proceedings of the 2009 *IEEE International Conference on Data Engineering, ICDE '09*. Washington, DC, USA: IEEE Computer Society, 2009, pp. 604–615.
- [23] E. Brill and R. C. Moore, “An improved error model for noisy channel spelling correction,” in Proceedings of the 38th Annual Meeting on *Association for Computational Linguistics*, ser. *ACL '00*. Morristown, NJ, USA: Association for Computational Linguistics, 2000, pp. 286–293.
- [24] M. Dreyer, J. R. Smith, and J. Eisner, “Latent-variable modeling of string transductions with finite-state methods,” in Proceedings of the Conference on *Empirical Methods in Natural Language Processing*, ser. *EMNLP '08*. Stroudsburg, PA, USA: Association for Computational Linguistics, 2008, pp. 1080–1089.